

# AMAT 19

## Finite Mathematics

### Valid Arguments



# Objectives

1. distinguish a valid argument from an invalid one
2. prove the validity of an argument using the rules of inference

## Terminologies

### *An argument*

is a sequence of compound propositions (premises are connected by a conjunction) that end with a conclusion<sup>1</sup>.

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<sup>1</sup>Usually the conclusion is preceded by the word “Therefore” (or something similar).

## Terminologies

### *An argument*

is a sequence of compound propositions (premises are connected by a conjunction) that end with a conclusion<sup>1</sup>.

### Example:

If Liza has a current password, then she can log onto the network.

Liza has a current password.

Therefore, Liza can log onto the network.

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is a sequence of compound propositions (premises are connected by a conjunction) that end with a conclusion<sup>1</sup>.

### Example:

If Liza has a current password, then she can log onto the network.

Liza has a current password. ← ↑ Premises/Hypothesis

Therefore, Liza can log onto the network. ← Conclusion

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## Terminologies

### *An argument*

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### Example:

If Liza has a current password, then she can log onto the network.

Liza has a current password.  $\leftarrow \uparrow$  Premises/Hypothesis

Therefore, Liza can log onto the network.  $\leftarrow$  Conclusion

Let  $p$ : Liza has a current password, and  $q$ : Liza can log onto the network.

In symbols,

$p \rightarrow q$

$p$

$\therefore q$ .

The argument is written as  $(p \rightarrow q) \wedge p \rightarrow q$ .

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## Terminologies

### VALID Argument:

An argument is valid iff

$$H_1 \wedge H_2 \wedge \dots \wedge H_n \Rightarrow C.$$

That is  $H_1 \wedge H_2 \wedge \dots \wedge H_n \rightarrow C$  is a tautology.

( $H_1, H_2, \dots, H_n$  may be simple or compound propositions, which are the premises of the argument, and  $C$  is the conclusion.)

**An argument is valid if the truth of its premises guarantees the truth of its conclusion.**

(Recall the conditional statement  $p \rightarrow q$ . This conditional statement is false whenever  $p$  is true but  $q$  is false. Hence, for a conditional statement to be valid, we want the truth of the premises to imply the truth of its conclusion.)

It can be verified using truth tables or using the rules of inference/logical implications and/or even the laws of equivalence of tautological forms.

## Terminologies

### SOUND Argument:

If an argument is VALID<sup>2</sup> and its premises are true, the argument is SOUND.

If an argument is not sound then it is UNSOUND.<sup>3</sup>

An argument can be valid even if its premises are false<sup>4</sup> - but such an argument is UNSOUND.

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<sup>2</sup>The conditional statement *premises*  $\rightarrow$  *conclusion* is a Tautology

<sup>3</sup>This is the negation of a sound argument. From this, can you tell me what is an unsound argument? Answer: An unsound argument means an argument is invalid or its premises are false.

<sup>4</sup>Note that False premises  $\rightarrow$  True conclusion is True and False premises  $\rightarrow$  False conclusion is also True.

## Example: Sound Argument

Example of an UNSOUND argument:

All toasters are items made of gold.<sup>5</sup>

All items made of gold are time-travel devices.

Therefore, all toasters are time-travel devices.

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<sup>5</sup>Note that for quantifiers “All”, it is the same as the conditional statement, i.e., “All toasters are items made of gold” is equivalent to “If it is a toaster, then it is an item made of gold.”

## Example: Sound Argument

Example of an UNSOUND argument:

All toasters are items made of gold.<sup>5</sup>

All items made of gold are time-travel devices.

Therefore, all toasters are time-travel devices.

This is a valid argument because this conditional statement results to a tautology. However, the premises are false. Hence, it is an unsound argument.

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## Recall: Logical Implication/Rules of Inference

| <i>Name</i>                   | <i>Rule</i>                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------|
| <i>Addition</i>               | $p \Rightarrow (p \vee q)$                                                              |
| <i>Simplification</i>         | $(p \wedge q) \Rightarrow p$                                                            |
| <i>Modus Ponens</i>           | $[p \wedge (p \rightarrow q)] \Rightarrow q$                                            |
| <i>Modus Tollens</i>          | $[(p \rightarrow q) \wedge \sim q] \Rightarrow \sim p$                                  |
| <i>Disjunctive Syllogism</i>  | $[(p \vee q) \wedge \sim p] \Rightarrow q$                                              |
| <i>Hypothetical Syllogism</i> | $[(p \rightarrow q) \wedge (q \rightarrow r)] \Rightarrow (p \rightarrow r)$            |
| <i>Constructive Dilemma</i>   | $[(p \rightarrow q) \wedge (r \rightarrow s)] \wedge (p \vee r) \Rightarrow (q \vee s)$ |
| <i>Distractive Dilemma</i>    | $(p \rightarrow q) \Rightarrow [p \rightarrow (p \wedge q)]$                            |

Another Important Law: **Reflexive**

$$P \Leftrightarrow P$$

Which rule of inference/law of equivalence is used?

It is below freezing now.

Therefore, it is below freezing or raining now.

Which rule of inference/law of equivalence is used?

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This is valid by Addition.

Which rule of inference/law of equivalence is used?

It is below freezing now.

Therefore, it is below freezing or raining now.

This is valid by Addition.

If Jack has a crush on Lucy then he will glance at her 3 times.  
He did not glance at her 3 times.

Thus, he does not have a crush on Lucy.

Which rule of inference/law of equivalence is used?

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This is valid by Modus Tollens.

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If Jack has a crush on Lucy then he will glance at her 3 times.  
He did not glance at her 3 times.

Thus, he does not have a crush on Lucy.

This is valid by Modus Tollens.

If it rains today, then we will not play football today.

If we do not play football today, then we will play football tomorrow. Therefore, if it rains today, then we will play football tomorrow.

## Which rule of inference/law of equivalence is used?

It is below freezing now.

Therefore, it is below freezing or raining now.

This is valid by Addition.

If Jack has a crush on Lucy then he will glance at her 3 times.  
He did not glance at her 3 times.

Thus, he does not have a crush on Lucy.

This is valid by Modus Tollens.

If it rains today, then we will not play football today.

If we do not play football today, then we will play football tomorrow. Therefore, if it rains today, then we will play football tomorrow.

This is valid by Hypothetical Syllogism. Why?

## Examples

Leah is going to school or to lunch with Anna.

If Leah is going to school, then it is not sunny.

It is sunny.

Is Leah going to lunch with Anna?

## Examples

Leah is going to school or to lunch with Anna.

If Leah is going to school, then it is not sunny.

It is sunny.

Is Leah going to lunch with Anna?

Note that the last statement is a question. This means we need to transform the last statement into a mathematical statement to argue whether Leah is going to lunch with Anna or not. We can form this into an argument by replacing the last statement with “Therefore, Leah is going to lunch with Anna”.

## Examples

Leah is going to school or to lunch with Anna.

If Leah is going to school, then it is not sunny.

It is sunny.

Is Leah going to lunch with Anna?

Note that the last statement is a question. This means we need to transform the last statement into a mathematical statement to argue whether Leah is going to lunch with Anna or not. We can form this into an argument by replacing the last statement with “Therefore, Leah is going to lunch with Anna”.

Now, let  $p$ : Leah is going to school,  $q$ : Leah is going to lunch with Anna, and  $r$ : It is sunny.

In symbols, the argument is written as  $(p \vee q) \wedge (p \rightarrow \sim r) \wedge r \rightarrow q$ .

## Examples

We have to prove the following statement:

$$(p \vee q) \wedge (p \rightarrow \sim r) \wedge r \rightarrow q.$$

We will use rules of inference/laws of equivalence of tautological forms to prove the above statement. Keep in mind that we need to show that the conditional statement results to a Tautology or a Value of True at the end.

Proof.

| Mathematical Statement                                                            | Reason                       |
|-----------------------------------------------------------------------------------|------------------------------|
| $(p \vee q) \wedge (p \rightarrow \sim r) \wedge r \rightarrow q$                 | Given                        |
| $(p \vee q) \wedge (p \rightarrow (\sim r)) \wedge \sim (\sim r) \rightarrow q$   | Double Negation              |
| $(p \vee q) \wedge [(p \rightarrow (\sim r)) \wedge \sim (\sim r)] \rightarrow q$ | Associativity of Conjunction |
| $(p \vee q) \wedge \sim p \rightarrow q$                                          | Modus Tollens                |
| $q \rightarrow q$                                                                 | Disjunctive Syllogism        |
| <b>T</b>                                                                          | Reflexive                    |



Therefore, basing from the established argument above, Leah is going to lunch with Anna.

As an exercise, use truth tables to prove that the above argument indeed results to a tautology.

## Examples

Mark will study mathematics or economics.

If Mark has to take Japanese, then he does not study economics.

Mark does not study mathematics.

Prove that Mark does not have to take Japanese.

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If Mark has to take Japanese, then he does not study economics.

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Prove that Mark does not have to take Japanese.

Note that to prove that Mark does not have to take Japanese, we need to replace the last statement with “Therefore, Mark does not have to take Japanese”.

## Examples

Mark will study mathematics or economics.

If Mark has to take Japanese, then he does not study economics.

Mark does not study mathematics.

Prove that Mark does not have to take Japanese.

Note that to prove that Mark does not have to take Japanese, we need to replace the last statement with “Therefore, Mark does not have to take Japanese”.

Let  $p$ : Mark will study Mathematics ,  $q$ : Mark will study Economics, and  $r$ : Mark will take Japanese.

In symbols, the argument is written as

$$(p \vee q) \wedge (r \rightarrow \sim q) \wedge \sim p \rightarrow \sim r.$$

## Examples

We have to prove the following statement:

$$(p \vee q) \wedge (r \rightarrow \sim q) \wedge \sim p \rightarrow \sim r.$$

We will use rules of inference/laws of equivalence of tautological forms to prove the above statement. Again, we need to show that the conditional statement results to a Tautology or a Value of True at the end. You may also verify using truth tables.

Proof.

| Mathematical Statement                                                        | Reason                                                                |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| $(p \vee q) \wedge (r \rightarrow \sim q) \wedge \sim p \rightarrow \sim r$   | Given                                                                 |
| $(p \vee q) \wedge \sim p \wedge (r \rightarrow \sim q) \rightarrow \sim r$   | Commutativity of Conjunction                                          |
| $((p \vee q) \wedge \sim p) \wedge (r \rightarrow \sim q) \rightarrow \sim r$ | Associativity of Conjunction or<br>Order of Precedence of Connectives |
| $q \wedge (r \rightarrow \sim q) \rightarrow \sim r$                          | Disjunctive Syllogism                                                 |
| $(r \rightarrow \sim q) \wedge q \rightarrow \sim r$                          | Commutativity of Conjunction                                          |
| $(r \rightarrow \sim q) \wedge \sim (\sim q) \rightarrow \sim r$              | Double Negation                                                       |
| $\sim r \rightarrow \sim r$                                                   | Modus Tollens                                                         |
| <b>T</b>                                                                      | Reflexive                                                             |



Therefore, basing from the established argument above, Mark does not have to take Japanese.

If Paul has a crush on Niki, then he will glance at her 3 times.

Paul glances at Niki 3 times.

Can we say that Paul has a crush on Niki?

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Paul glances at Niki 3 times.

Can we say that Paul has a crush on Niki?

Note that to prove this, replace the last statement with  
“Therefore, Paul has a crush on Niki”.

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Paul glances at Niki 3 times.

Can we say that Paul has a crush on Niki?

Note that to prove this, replace the last statement with  
“Therefore, Paul has a crush on Niki”.

Let  $p$ : Paul has a crush on Niki, and  $q$ : Paul glances at Niki 3 times.

In symbols, the argument is written as  $(p \rightarrow q) \wedge q \rightarrow p$ .

If Paul has a crush on Niki, then he will glance at her 3 times.

Paul glances at Niki 3 times.

Can we say that Paul has a crush on Niki?

Note that to prove this, replace the last statement with  
“Therefore, Paul has a crush on Niki”.

Let  $p$ : Paul has a crush on Niki, and  $q$ : Paul glances at Niki 3 times.

In symbols, the argument is written as  $(p \rightarrow q) \wedge q \rightarrow p$ .

Whatever rules of inference or laws of equivalence you will use here will not give you any valid conclusion. If you try to use truth tables, you will find out that this does not result to a Tautology. This is because this argument is INVALID! This is a common INVALID argument that has a name: FALLACY OF THE CONVERSE.

All dogs are hairy.

My pet Kabri is not a dog.

Therefore, Kabri is not hairy.

All dogs are hairy.

My pet Kabri is not a dog.

Therefore, Kabri is not hairy.

Again, “All dogs are hairy” is equivalent to the statement “If it is a dog, then it is hairy”.

All dogs are hairy.

My pet Kabri is not a dog.

Therefore, Kabri is not hairy.

Again, “All dogs are hairy” is equivalent to the statement “If it is a dog, then it is hairy”.

Let  $p$ : It is a dog, and  $q$ : It is hairy. Note that “IT” can be replaced by any name such as “Kabri”.

In symbols, the argument is written as  $(p \rightarrow q) \wedge \sim p \rightarrow \sim q$ .

All dogs are hairy.

My pet Kabri is not a dog.

Therefore, Kabri is not hairy.

Again, “All dogs are hairy” is equivalent to the statement “If it is a dog, then it is hairy”.

Let  $p$ : It is a dog, and  $q$ : It is hairy. Note that “IT” can be replaced by any name such as “Kabri”.

In symbols, the argument is written as  $(p \rightarrow q) \wedge \sim p \rightarrow \sim q$ .

Similar to the previous example, whatever rules of inference or laws of equivalence you will be using here will not give you any valid conclusion. Using truth tables, you will find out that this also does not result to a Tautology. This is also an INVALID argument known as the FALLACY OF THE INVERSE.

# Fallacy

Formal FALLACIES are errors that result from misapplying or not following these rules.

These are incorrect reasoning that leads to invalid arguments.

## Common Formal Fallacies

Here are some common formal fallacies. Try to use truth tables to prove that they will not result to a Tautology.

1. P or Q. Therefore, P.
2. *Affirming the Disjunct*: P or Q. P. Therefore, not Q.
3. *Denying the Conjunct*: Not both P and Q are true. Not P. Therefore, Q.
4. *Affirming the Consequent (Fallacy of the Converse)*: If P then Q. Q. Therefore, P.
5. *Denying the Antecedent (Fallacy of the Inverse)*: If P then Q. Not P. Therefore, not Q.
6. *Non Sequitur of Evidence*: If P then Q. R. Therefore, Q.
7. *Non Sequitur of Evidence*: If P then Q. Not R. Therefore, not P.
8. *Non Sequitur of Relevance*: If P then Q. P. Therefore, R.
9. *Non Sequitur of Relevance*: If P then Q. Not Q. Therefore, not R.